Fall 2025 EE582 Homework 02 Report
Nodal-Analysis-based Modeling of Electric Circuits

Aryan Ritwajeet Jha

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Contents

Section A (Numerical Oscillations of Discretization Rules): Question 1

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at $t = 0$ and the voltage source takes its value at $t = \Delta t$.

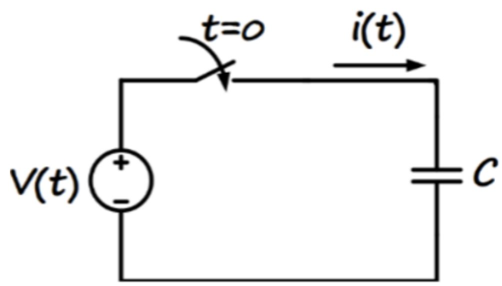

Figure 1: Simple capacitor circuit with DC voltage source

1) Determine the step-by-step solution for the current in the capacitor (in terms of capacitance C, time-step size Δt, and source voltage) using the following discretization rules: a) Forward Euler, b) Backward Euler, c) Trapezoidal.

Trapezoidal Discretization of Capacitor

The equations for the Trapezoidal discretization of a capacitor are (see Appendix I for derivation):

The final equations for $v(t)$ and $i(t)$ are as follows:

$$\boxed{v(t) = e_h + R_C i(t)} \tag{1}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{2}$$

where:

- $R_C = \frac{\Delta t}{2C}$ is the equivalent resistance.
- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$ is the history voltage.
- $i_h = \frac{v(t-\Delta t)}{R_C} - i(t - \Delta t)$ is the history current.

Step-by-Step Solution Table for Trapezoidal Discretization

Table 1 shows the step-by-step evolution of the system variables based on trapezoidal discretization.

t	$v(t)$	$i(t)$	e_h	i_h
-h	V_0	0	–	–
0^+	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$	V_0	$\frac{V_0}{R_C}$
h	V_{DC}	$-\frac{V_{DC}}{R_C} + \frac{V_0}{R_C}$	$2V_{DC} - V_0$	$\frac{2V_{DC}}{R_C} - \frac{V_0}{R_C}$
$2h$	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$	V_0	$\frac{V_0}{R_C}$
$3h$	V_{DC}	$-\frac{V_{DC}}{R_C} + \frac{V_0}{R_C}$	$2V_{DC} - V_0$	$\frac{2V_{DC}}{R_C} - \frac{V_0}{R_C}$
$4h$	V_{DC}	$\frac{V_{DC}}{R_C} - \frac{V_0}{R_C}$	V_0	$\frac{V_0}{R_C}$

Table 1: Step-by-step solution for trapezoidal discretization of circuit in Figure 1

Backward-Euler (BE) Discretization of Capacitor

The equations for the Backward-Euler discretization of a capacitor are (see Appendix II for derivation):

$$\boxed{v(t) = e_h + R_C\, i(t)} \tag{3}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{4}$$

where:

- $R_C = \dfrac{\Delta t}{C}$
- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$

- $i_h = \dfrac{v(t-\Delta t)}{R_C}$

Step-by-Step Solution Table for Backward-Euler Discretization

Table 2 shows the step-by-step evolution of the system variables based on backward euler discretization.

t	$v(t)$	$i(t)$	e_h	i_h
$-h$	V_0	0	$-$	$-$
0^+	V_{DC}	$\dfrac{V_{DC}}{R_C} - \dfrac{V_0}{R_C}$	V_0	$\dfrac{V_0}{R_C}$
h	V_{DC}	0	V_{DC}	$\dfrac{V_0}{R_C}$
$2h$	V_{DC}	0	V_{DC}	$\dfrac{V_{DC}}{R_C}$
$3h$	V_{DC}	0	V_{DC}	$\dfrac{V_{DC}}{R_C}$
$4h$	V_{DC}	0	V_{DC}	$\dfrac{V_{DC}}{R_C}$

Table 2: Step-by-step solution for backward euler discretization of circuit in Figure 1

Forward Euler (FE) Discretization of Capacitor

Upon solving for the forward euler discretization of the capacitor, we encounter the issue of division by zero as $v(t)$ and $e_h(t)$ are equal at every time step (one source connected to another source directly), and $i(t)$ becomes undefined (because of lack of any resistance between them). $i(t)$ could be of an arbitrarily high magnitude, however in order to ascertain its sign, we introduce a very small resistance $R_\epsilon \ll R_c$ in series with the capacitor to avoid division by zero. This resistance is artificial and does not represent any physical element in the circuit (or at best could be argued to represent the relatively negligible resistance of the connecting wires).

The equations for the Forward Euler discretization of a capacitor are (see Appendix III for derivation):

$$\boxed{v(t) = e_h(t)} \tag{5}$$

$$\boxed{i(t) = \frac{v(t)}{R_\epsilon} - i_h(t)} \tag{6}$$

where:

- $R_c = \dfrac{h}{C}$ (discretized capacitor resistance)
- $R_\epsilon \ll \dfrac{h}{C}$ (artificial resistance for FE table)
- $e_h(t) = v(t-h) + R_c i(t-h)$
- $i_h(t) = \frac{e_h(t)}{R_\epsilon}$

Step-by-Step Solution Table for Forward-Euler Discretization

Table 3 shows the step-by-step evolution of the system variables based on forward euler discretization, assuming a small resistance R_ϵ to avoid division by zero.

t	$v(t)$	$i(t)$	$e_h(t)$	$i_h(t)$
$-h$	V_0	0	$-$	$-$
0^+	V_{DC}	$\dfrac{V_{DC} - V_0}{R_\epsilon}$	V_0	$\dfrac{V_0}{R_\epsilon}$
h	V_{DC}	$\dfrac{V_0 - V_{DC}}{R_\epsilon^2 / R_c}$	$V_{DC}\left(1 + \frac{R_c}{R_\epsilon}\right) - V_0\left(\frac{R_c}{R_\epsilon}\right)$	$\dfrac{V_{DC}\left(1 + \frac{R_c}{R_\epsilon}\right) - V_0\left(\frac{R_c}{R_\epsilon}\right)}{R_\epsilon}$
$2h$	V_{DC}	$\dfrac{V_{DC} - V_0}{R_\epsilon^3 / R_c^2}$	$-V_{DC}\left(\frac{R_c^2}{R_\epsilon^2} - 1\right) + V_0\left(\frac{R_c^2}{R_\epsilon^2}\right)$	$\dfrac{-V_{DC}\left(\frac{R_c^2}{R_\epsilon^2} - 1\right) + V_0\left(\frac{R_c^2}{R_\epsilon^2}\right)}{R_\epsilon}$
$3h$	V_{DC}	$\dfrac{V_0 - V_{DC}}{R_\epsilon^4 / R_c^3}$	$V_{DC}\left(1 + \frac{R_c^3}{R_\epsilon^3}\right) - V_0\left(\frac{R_c^3}{R_\epsilon^3}\right)$	$\dfrac{V_{DC}\left(1 + \frac{R_c^3}{R_\epsilon^3}\right) - V_0\left(\frac{R_c^3}{R_\epsilon^3}\right)}{R_\epsilon}$

Table 3: Step-by-step solution for forward euler discretization of circuit in Figure 1. Note the diverging oscillations with successive powers of (R_c/R_ϵ), demonstrating numerical instability.

Section A: Question 2

Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

Section B (Basic Circuit Numerical Simulation): Question 1

The parameters of the circuit below are: $R = 10\Omega$, $L = 20$mH, $e(t) = 10$V dc, and the switch is closed at $t = 0$. The circuit below is to be solved from $t = 0$ to $t = 10$ms using: a) PSCAD software, b) your own MATLAB code based on the nodal analysis method.

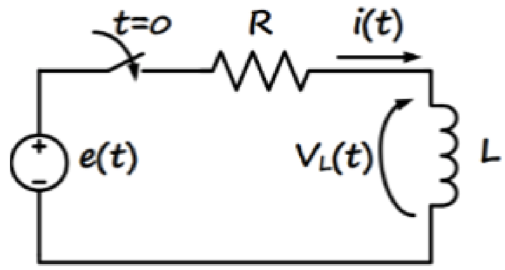

Figure 2: RL circuit with DC voltage source and switch

1) Obtain the solution for the inductor current and voltage using PSCAD with time-step sizes $\Delta t = 0.1$ms and $\Delta t = 0.8$ms.

Section B : Question 2

2) Discretize the circuit and write a MATLAB code to solve the circuit step-by-step using the two discretization rules: a) Backward Euler, b) Trapezoidal, each with time-step sizes $\Delta t = 0.1\text{ms}$ and $\Delta t = 0.8\text{ms}$.

Section B: Question 3

3) Show the results for $i(t)$ obtained in the six cases (PSCAD, your Backward Euler code, and your Trapezoidal code; each with two different step sizes) superimposed in a (big enough) Plot. Show the same for $v_L(t)$ in a separate plot. Comment on the accuracy of the methods, impact of the time-step size, and how the PSCAD results compare with your Trapezoidal code.

Section C (Circuit Interruption): Question 1

The case below (figure on the left) represents a typical short-circuit fault in transmission lines and the breaker operation for line interruption. The equivalent circuit is shown in the figure on the right.

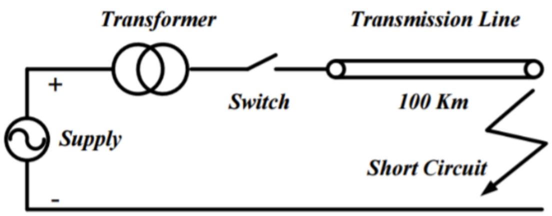

Figure 3: Transmission line with short-circuit fault

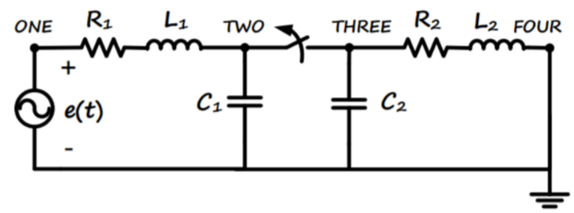

Figure 4: Equivalent circuit representation

On the left-hand side of the switch/breaker, we have the source voltage and a resistance and inductance representing the supply system combined with the transformer. The capacitor to the left of the breaker represents the combined substation capacitance due to the substation busbars. On the right-hand side of the breaker, we have the transmission line parameters: resistance, inductance, and capacitance represented as simple lumped parameters, where the right side is short-circuited due to the fault, assuming that the fault has already been applied. We have assumed a single-phase circuit here for simplicity. The parameters of the circuit below are: $e(t) = 230 \times \cos(\omega t)$ kV, $\omega = 377$ rad/s, $R_1 = 3\Omega$, $L_1 = 350$ mH, $R_2 = 50\Omega$, $L_2 = 100$ mH, $C_1 = 10$ nF, $C_2 = 600$ nF. Assume that the system is operating with zero initial conditions and the breaker is initially closed. At $t = 8$ ms, the breaker is commanded to open the line, but it will not actually open until the current passes through zero. The study period is from $t = 0$ to $t = 20$ ms.

1) Choose an appropriate simulation time-step size Δt (assuming the Trapezoidal integration rule) according to the time constants in the circuit. For this, evaluate the approximate resonant frequencies analytically before and after the breaker operation.

Section C: Question 2

2) Write your own MATLAB code to solve the system and also obtain the solution using PSCAD.

Section C: Question 3

3) Show the results for voltages at nodes Two and $Three$ and the fault current (i.e., from node $Four$ to $ground$) in three separate (big enough) plots. In each plot, superimpose the results from your code and also from PSCAD, both obtained using the same simulation time-step size obtained in part 1) for comparison. Comment on the accuracy of your results compared to those of PSCAD.

I Appendix: Derivation of Trapezoidal Discretization for Capacitor

To derive the equations v_t and i_t, the following steps are involved:

- Start with the governing equation for a capacitor:

$$i(t) = C\frac{dv(t)}{dt} \tag{7}$$

- Apply the trapezoidal rule to discretize the relationship between voltage and current:

$$v(t) - v(t - \Delta t) = \frac{\Delta t}{2C}\left[i(t) + i(t - \Delta t)\right] \tag{8}$$

- Define the equivalent resistance R_C and history terms:

$$R_C = \frac{\Delta t}{2C} \tag{9}$$

- Derive the Thevenin (series) and Norton (parallel) companion models.

The final equations for the capacitor are:

$$\boxed{v(t) = e_h + R_C i(t)} \tag{10}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{11}$$

where:

- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$
- $i_h = \frac{v(t-\Delta t)}{R_C} - i(t - \Delta t)$

II Appendix: Derivation of Backward-Euler Discretization for Capacitor

Starting from the capacitor constitutive relation

$$i(t) = C\frac{dv(t)}{dt}, \tag{12}$$

the Backward-Euler (BE) discretization at time t approximates the derivative by

$$\frac{dv(t)}{dt} \approx \frac{v(t) - v(t-\Delta t)}{\Delta t}. \tag{13}$$

Substituting and rearranging yields

$$i(t) \approx C\frac{v(t) - v(t-\Delta t)}{\Delta t}. \tag{14}$$

Writing the BE companion in a Thévenin/Norton form gives an equivalent resistance and history terms. The final boxed companion equations used in the main text (capacitor, BE) are

$$\boxed{v(t) = e_h + R_C\, i(t)} \tag{15}$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \tag{16}$$

with the BE companion parameters (for the capacitor):

$$R_C = \frac{\Delta t}{C}, \tag{17}$$

$$e_h = v(t-\Delta t), \tag{18}$$

$$i_h = \frac{v(t-\Delta t)}{R_C}. \tag{19}$$

These expressions are used when implementing the Backward-Euler discretization in the numerical examples and the BE table in the main text.

III Appendix: Derivation of Forward-Euler Discretization for Capacitor

Starting from the capacitor constitutive relation

$$i(t) = C\frac{dv(t)}{dt}, \tag{20}$$

the Forward-Euler (FE) discretization at time t approximates the derivative using the previous time step:

$$\frac{dv(t)}{dt} \approx \frac{v(t) - v(t-h)}{h}. \tag{21}$$

Substituting and rearranging yields

$$i(t) \approx C\frac{v(t) - v(t-h)}{h}. \tag{22}$$

To cast this in a companion model form similar to BE and trapezoidal methods, we introduce an artificial small resistance $R_\epsilon = h/C$ (as justified in the main text to avoid division by zero). The final boxed companion equations used in the main text (capacitor, FE) are

$$\boxed{v(t) = e_h(t)} \tag{23}$$

$$\boxed{i(t) = \frac{v(t) - e_h(t)}{R_\epsilon}} \tag{24}$$

with the FE companion parameters (for the capacitor):

$$R_c = \frac{h}{C} \quad , \tag{25}$$

$$e_h(t) = v(t-h) + R_c\, i(t-h), \tag{26}$$

$$i_h(t) = \frac{e_h(t)}{R_c}. \tag{27}$$

These expressions are used when implementing the Forward-Euler discretization in the numerical examples and the FE table in the main text.